

DSA Practical No: 05

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Stack Using Array :

INPUT :

```
#include <stdio.h>

#include <stdlib.h>

#define SIZE 5

struct stack
{
    int s[SIZE];
    int top;
};

int isFull(struct stack *st)
{
    if (st->top >= SIZE - 1)
        return 1;
    else
        return 0;
}

int isEmpty(struct stack *st)
{
    if (st->top == -1)
        return 1;
```

```

        else
            return 0;
    }

void push(struct stack *st, int item)
{
    st->top++;
    st->s[st->top] = item;
}

void pop(struct stack *st)
{
    int item;

    if (isEmpty(st))
    {
        printf("\nStack is empty! Underflow.\n");
    }
    else
    {
        item = st->s[st->top];
        st->top--;

        printf("\nThe popped element is: %d\n", item);
    }
}

void display(struct stack *st)

```

```
{  
    int i;  
  
    if (isEmpty(st))  
    {  
        printf("\nStack is empty!\n");  
    }  
    else  
    {  
        printf("\nStack elements are:\n");  
  
        for (i = st->top; i >= 0; i--)  
        {  
            printf("%d\n", st->s[i]);  
        }  
  
        printf("TOP = %d\n", st->top);  
    }  
}
```

```
int main()  
{  
    struct stack st;  
    int choice;  
    int item;  
    char ans;  
  
    st.top = -1;
```

```
do
{

    printf("IMPLEMENTATION OF STACK\n");
    printf("1. PUSH\n");
    printf("2. POP\n");
    printf("3. DISPLAY\n");
    printf("4. EXIT\n");

    printf("\nEnter your choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            printf("\nEnter the item to be pushed: ");
            scanf("%d", &item);

            if (isFull(&st))
            {
                printf("\nStack is full!\n");
            }
            else
            {
                push(&st, item);
                printf("\nItem %d pushed successfully.\n", item);
            }
        }
```

```
break;
```

```
case 2:
```

```
pop(&st);
```

```
break;
```

```
case 3:
```

```
display(&st);
```

```
break;
```

```
case 4:
```

```
printf("\nExiting program...\n");
```

```
exit(0);
```

```
default:
```

```
printf("\nInvalid choice!\n");
```

```
}
```

```
printf("\nDo you want to continue? (y/n): ");
```

```
scanf(" %c", &ans);
```

```
} while (ans == 'y' || ans == 'Y');
```

```
printf("\nProgram ended.\n");
```

```
return 0;
```

```
}
```

OUTPUT :

```
PS D:\dsa\Experiment-05-Stack-and-Queue> .\stack.exe
IMPLEMENTATION OF STACK
1. PUSH
2. POP
3. DISPLAY
4. EXIT

Enter your choice: 1

Enter the item to be pushed: 12

Item 12 pushed successfully.

Do you want to continue? (y/n): y
IMPLEMENTATION OF STACK
1. PUSH
2. POP
3. DISPLAY
4. EXIT

Enter your choice: 1

Enter the item to be pushed: 13

Item 13 pushed successfully.
```

```
Do you want to continue? (y/n): y
IMPLEMENTATION OF STACK
1. PUSH
2. POP
3. DISPLAY
4. EXIT

Enter your choice: 1

Enter the item to be pushed: 14

Item 14 pushed successfully.

Do you want to continue? (y/n): y
IMPLEMENTATION OF STACK
1. PUSH
2. POP
3. DISPLAY
4. EXIT

Enter your choice:
1

Enter the item to be pushed: 15

Item 15 pushed successfully.
```

```
Do you want to continue? (y/n): y
IMPLEMENTATION OF STACK
1. PUSH
2. POP
3. DISPLAY
4. EXIT
```

```
Enter your choice: 2
```

```
The popped element is: 15
```

```
Do you want to continue? (y/n): y
IMPLEMENTATION OF STACK
1. PUSH
2. POP
3. DISPLAY
4. EXIT
```

```
Enter your choice: 3
```

```
Stack elements are:
14
13
12
TOP = 2
```

```
Do you want to continue? (y/n): y
```

```
Do you want to continue? (y/n): y
IMPLEMENTATION OF STACK
1. PUSH
2. POP
3. DISPLAY
4. EXIT
```

```
Enter your choice: 4
```

```
Exiting program...
```

```
PS D:\dsa\Experiment-05-Stack-and-Queue>
```